

Project Initialization and Planning Phase

Date	30 September 2026
Team ID	PNT2026TMID98765
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform comic story creation using Google Gemini models, boosting authoring efficiency, visual story consistency, and audience engagement. It tackles traditional illustration bottlenecks, promising automated script-to-panel rendering, context-aware character persistence, and interactive digital publishing. Key features include Gemini multimodal script parsing, sequential panel layout generation, and real-time dialogue bubble synthesis.

Project Overview	
Objective	The primary objective is to build ComicCraft, an end-to-end AI comic creation platform using Gemini models to automate script parsing, panel layout, visual art prompting, and dialogue placement.
Scope	The project comprehensively covers story idea expansion, multimodal script breakdown, character visual memory persistence across panels, speech bubble overlay, and digital comic PDF/web export.
Problem Statement	
Description	Traditional comic production requires expert illustration skills, extensive drawing time (10–20 hrs/page), and high art budgets, creating severe entry barriers for non-artist storytellers.
Impact	Solving these issues democratizes visual storytelling, reduces comic production timelines by 90%, lowers costs, and enables rapid iteration of indie webtoons and educational graphic novels.
Proposed Solution	
Approach	Employing Google Gemini 1.5 Pro / Flash models for multimodal script understanding, character consistency tracking, automated visual prompt generation, and sequential panel layout synthesis.
Key Features	• Gemini Multimodal Script & Dialogue Parser. • Sequential Character Consistency & Style Anchor Engine. • Dynamic Multi-Panel Layout Generator with Auto-Speech Bubbles. • Real-time Interactive Creator Canvas & Web/PDF Viewer. • Continuous learning and story template adaptation.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	NVIDIA T4 / A10G GPU (Cloud/Local), 8 vCPU Cores
Memory	RAM specifications	16 GB System RAM, 16 GB VRAM
Storage	Disk space for data, models, and logs	500 GB High-Speed NVMe SSD
Software		
Frameworks	Python / Web frameworks	FastAPI, Streamlit / React, PyTorch, LangChain
Libraries	Additional libraries	google-generativeai SDK, Pillow, OpenCV, ReportLab, Pandas, NumPy
Development Environment	IDE	VS Code, Jupyter Notebook, Google Colab
Data		
Data	Source, size, format	Custom Comic Script & Panel Datasets, Gemini Multimodal Prompts (10,000+ panels, JSON/CSV)